Literature Review

The Clueless

Westcliff University

Data200: Applied Statistical Analytics

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Literature Review

Ahmad, B., Chen, J., & Chen, H. (2025). Feature selection strategies for optimized heart disease diagnosis using ML and DL models. *arXiv*. https://doi.org/10.48550/arXiv.2503.16577

This research paper examines the application of feature selection using ANOVA for predicting heart disease and the role of important variables like cholesterol, blood pressure, and age. This paper clearly shows how traditional statistical analysis can be used to decrease dimensionality and still maintain predictive ability. This paper stresses the importance of hypothesis testing, the size of effects, and the utilization of summary statistics for the interpretation of variable importance. This paper shows how exploratory data analysis can be combined with traditional statistical analysis. This source is directly relevant to the proposed project workflow.

Ramesh, R. S., Udupa, R. T. S., Monisha, J., & Kushi, K. K. S. (2025). Cardiovascular disease prediction using machine learning: A comparative analysis. *arXiv*. https://doi.org/10.48550/arXiv.2507.21898

In this paper, classic inferential statistics such as ANOVA, t-tests, Chi-squared tests, and logistic regression are employed to dig through a massive dataset of cardiovascular information in search of interesting predictors of heart disease. The paper emphasizes the importance of exploratory data analysis and effect size estimation to determine the relevance of risk factors to outcomes. By means of specific illustrations of hypothesis testing and p-value calculation in a real-world medical context, this paper illustrates the continued relevance of classic statistical techniques. This source is thus a useful methodological guide to performing high-quality statistical analysis on the Kaggle heart disease dataset.

Zulkiflee, N. F., & Rusiman, M. S. (2021). Heart Disease Prediction Using Logistic Regression. *Enhanced Knowledge in Sciences and Technology*, 1(2), 177-184. https://publisher.uthm.edu.my/periodicals/index.php/ekst/article/view/2163

In this study, logistic regression analysis is used to determine the important clinical risk factors for heart disease, such as age, cholesterol, blood pressure, type of chest pain, and number of involved major vessels. The authors highlight the significance of statistical significance and odds ratios in determining the strength of association. The approach taken in this study involves diagnosing the model to ensure that the assumptions are satisfied, which is consistent with the approach that will be taken in this project. This study is an excellent example of how classical statistical models, as opposed to machine learning models, can be used to discover important clinical insights.